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Sec - 28

Mat - 120

Fall - 23

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14.1

$$\textcircled{3} \int_2^4 \int_0^1 x^2 y \, dx \, dy$$

$$= \int_2^4 y \left[\frac{x^3}{3} \right]_0^1 dy$$

$$= \frac{1}{3} \int_2^4 y (1^3 - 0^3) dy$$

$$= \frac{1}{3} \int_2^4 y \, dy$$

$$= \frac{1}{3} \left[\frac{y^2}{2} \right]_2^4 = \frac{1}{6} (4^2 - 2^2)$$

$$= \frac{1}{6} (16 - 4) = \frac{12}{6} = 2$$

$$\textcircled{8} \int_4^6 \int_3^7 dy \, dx$$

$$= \int_4^6 [y]_3^7 dx$$

$$= \int_4^6 10 \, dx$$

$$= 10 [x]_4^6 = 10(6 - 4) = 20 \text{ (Ans)}$$

$$\textcircled{22} \int_3^4 \int_3^2 \frac{1}{(x+y)^2} dy \, dx$$

$$= \int_3^4 \left[\frac{(x+y)^{-2+1}}{-2+1} \right]_3^2 dx$$

$$= \int_3^4 \left[\frac{1}{x+y} \right]_3^2 dx$$

$$u = x+y$$

$$= - \int_3^4 \frac{1}{2+x} - \frac{1}{1+x} dx$$

$$= \int_3^4 \frac{1}{1+x} dx - \int_3^4 \frac{1}{x+2} dx$$

$$= (\cancel{\ln(x)} [\ln(1+x)]_3^4 - [\ln(x+2)]_3^4)$$

$$= \ln(4+1) - \ln(3+1) - \ln(4+2) + \ln(3+2)$$

$$= \ln(5) + \ln(5) - \{\ln(4) + \ln(6)\}$$

$$= 2 \ln(5) - \{\ln(4 \times 6)\}$$

$$= \ln(5^2) - \ln(24)$$

$$= \ln\left(\frac{25}{24}\right)$$



13

$$\iint_R 4xy^3 dA, \quad R = \{(x,y) : -1 \leq x \leq 1, -2 \leq y \leq 2\}$$

$$= 4 \int_{-2}^2 \int_{-1}^1 xy^3 dx dy$$

$$= 4 \int_{-2}^2 y^3 \left[\frac{x^2}{2} \right]_{-1}^1 dy$$

$$0.2 \int_{-2}^2 y^3 (1-y) dy$$

$$= 0 \text{ (Ans)}$$

[16]

$$\iint_R (x \sin y - y \sin x) dA, \quad R = \{(x, y) : 0 \leq x \leq \pi/2, 0 \leq y < \pi/3\}$$

$$= \int_0^{\pi/3} \int_0^{\pi/2} (x \sin y - y \sin x) dx dy$$

$$= \int_0^{\pi/3} \left[\int_0^{\pi/2} x \sin y dx - \int_0^{\pi/2} y \sin x dx \right] dy$$

$$= \int_0^{\pi/3} \left\{ \sin y \left[\frac{x^2}{2} \right]_0^{\pi/2} - y [-\cos x]_0^{\pi/2} \right\} dy$$

$$= \int_0^{\pi/3} \left\{ \sin y \left(\frac{\pi^2}{4 \times 2} - 0 \right) + y [\cos(\pi/2) - 0] \right\} dy$$

$$= \int_0^{\pi/3} \frac{\pi^2}{8} \sin y dy + \int_0^{\pi/3} y (0 - 1) dy$$

$$= -\frac{\pi}{8} [\cos y]_0^{\frac{\pi}{3}} - \left[\frac{y}{2} \right]_0^{\frac{\pi}{3}} = -\frac{\pi}{8} \times \left(\frac{1}{2} - 1 \right) - \left(\frac{\pi}{6} - 0 \right)$$

$$\Rightarrow \frac{\pi}{16} - \frac{\pi}{18} =$$

$$= \frac{\pi}{144}$$

Q11

$$\text{Ans: } (x+y)^2 = x^2 + y^2 + 2xy$$

$$\text{Differentiating both sides w.r.t. } x$$

$$2(x+y) \cdot 1 = 2x + 2y$$

$$2(x+y) = 2x + 2y$$

$$2(x+y) = 2x + 2y$$

$$2(x+y) = 2x + 2y$$

14.2

$$\boxed{9} \int_{\frac{1}{4}}^1 \int_{x^2}^x \sqrt{\frac{x}{y}} dy dx$$

$$\Rightarrow \int_{\frac{1}{4}}^1 \int_{x^2}^x x^{\frac{1}{2}} \cdot y^{-\frac{1}{2}} dy dx = \int_{\frac{1}{4}}^1 x^{\frac{1}{2}} \int_{x^2}^x y^{-\frac{1}{2}} dy dx$$

$$= \int_{\frac{1}{4}}^1 x^{\frac{1}{2}} \left[\frac{y^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} \right]_{x^2}^x dx$$

$$= 2 \int_{\frac{1}{4}}^1 x^{\frac{1}{2}} \left[y^{\frac{1}{2}} \right]_{x^2}^x dx \Rightarrow 2 \int_{\frac{1}{4}}^1 x^{\frac{1}{2}} (x^{\frac{1}{2}} - x) dx$$

$$= 2 \int_{\frac{1}{4}}^1 (x - x^{\frac{3}{2}}) dx$$

$$\Rightarrow 2 \int_{\frac{1}{4}}^1 x dx - 2 \int_{\frac{1}{4}}^1 x^{\frac{3}{2}} dx$$

$$\begin{aligned}
&= 2 \left[\frac{x^2}{2} \right]_{\frac{1}{4}}^1 - 2 \left[\frac{x^{\frac{3}{2}+1}}{\frac{3}{2}+1} \right]_{\frac{1}{4}}^1 \\
&\Rightarrow 2 \left[\frac{1}{2} - \frac{1}{4^2 \times 2} \right] - 2 \left[\frac{x^{\frac{5}{2}}}{\frac{5}{2}} \right]_{\frac{1}{4}}^1 \\
&\Rightarrow 2 \left[\frac{1}{2} - \frac{1}{16 \times 2} \right] - 2 \times \frac{2}{5} \left[\left(\frac{1}{4} \right)^{\frac{5}{2}} - \left(\frac{1}{4} \right)^{\frac{5}{2}} \right] \\
&= 2 \left(\frac{15}{32} \right) - \frac{4}{5} \left(1 - \frac{1}{32} \right) \\
&\Rightarrow \frac{15}{16} - \frac{4}{5} \left(\frac{31}{32} \right) \\
&\Rightarrow \frac{15}{16} - \frac{31}{40} \Rightarrow \frac{13}{80} \text{ (Ans)}
\end{aligned}$$

[7]

$$\begin{aligned}
&\int_0^1 \int_0^x y \sqrt{x^2 - y^2} dy dx \\
&= \int_0^1 \left(\sqrt{x^2 - y^2} \int_0^x y dy \right) dx \\
&= \int_0^1 \sqrt{x^2 - y^2} \left[\frac{y^2}{2} \right]_0^x dx
\end{aligned}$$

(7)

[7]

$$\int_0^1 \int_0^1 y \sqrt{x^2 - y^2} \, dy \, dx$$

$$\int_0^1 \int_0^x \sqrt{x^2 - y^2} \, dy \, dx$$

$$\int_0^1 \int_0^1 xy - y^2 \, dy \, dx$$

$$\int_0^1 \int_0^x xy \, dy - \int_0^1 y^2 \, dy \, dx$$

$$\int_0^1 x \left[\frac{y^2}{2} \right]_0^x - \left[\frac{y^3}{3} \right]_0^x \, dx$$

$$\int_0^1 x \left(\frac{x^2}{2} - 0 \right) - \left(\frac{x^3}{3} \right) \, dx$$

$$\int_0^1$$

$$x^3 \, dx$$

$$\Rightarrow \frac{1}{2} \int_0^1 x^3 \, dx - \frac{1}{3} \int_0^1$$

$$\Rightarrow \frac{1}{2} \left[\frac{x^4}{4} \right]_0^1 - \frac{1}{3} \left[\frac{x^4}{4} \right]_0^1$$

$$\Rightarrow \frac{1}{8} - \frac{1}{12} \Rightarrow \frac{12-8}{8 \times 12}$$

$$\Rightarrow \frac{x}{8 \times 12} = \frac{4}{12}$$

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19

$\iint_R x(1+y)^{-\frac{1}{2}} \frac{1}{2} dA$, R is the region in the first quadrant enclosed by $y = x^2$, $y = 4$ and $x = 0$.

$$= \int_0^4 \int_0^{\sqrt{y}} x(1+y)^{-\frac{1}{2}} dx dy$$

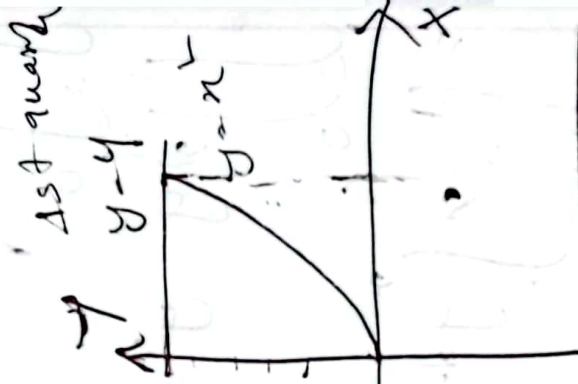
$$= \int_0^4 \int_0^{\sqrt{y}} x(1+y)^{-\frac{1}{2}} dx dy$$

$$= \int_0^4 (1+y)^{-\frac{1}{2}} \left[\frac{x^2}{2} \right]_0^{\sqrt{y}} dy$$

$$= \int_0^4 (1+y)^{-\frac{1}{2}} \frac{1}{2} \left[\frac{x^2}{2} \right]_0^{\sqrt{y}} dy$$

$$= \int_0^4 (1+y)^{-\frac{1}{2}} \left(\frac{y}{2} \right) dy$$

$$= \frac{1}{2} \int_0^4 y(1+y)^{-\frac{1}{2}} dy$$



$$\begin{aligned} x &= y \\ x &= \pm 2 \\ 0 &\leq x \leq 2 \\ x^2 &\leq y \leq 4 \end{aligned}$$

⑨

$$= \frac{1}{2} \int_{-1}^{17} (u-1)^{-\frac{1}{2}} du$$

$$= \frac{1}{2} \left[\frac{(u-1)^{\frac{1}{2}}}{\frac{1}{2}} \right]_{-1}^{17}$$

$$= \frac{1}{4} \times 2 \left[\sqrt{17}-1 \right] = \frac{1}{2} (\sqrt{17}-1)$$

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$$xy=1 \Rightarrow y = \frac{1}{x} \quad \text{--- (i)}$$

$$x = y \quad \text{--- (ii)}$$

$$y = 2x \quad \text{--- (iii)}$$

from (i) and (ii)

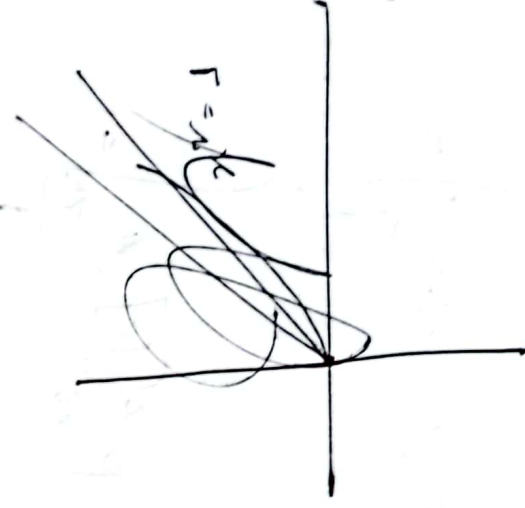
$$\frac{1}{x} = x$$

$$x^2 = 1$$

$$x = \pm 1$$

$$0 < x \leq 1$$

$$\begin{array}{l} y = \frac{1}{x} \\ 2x = \frac{1}{x} \\ x^2 = \frac{1}{2} \\ \boxed{x = \frac{1}{\sqrt{2}}} \end{array}$$



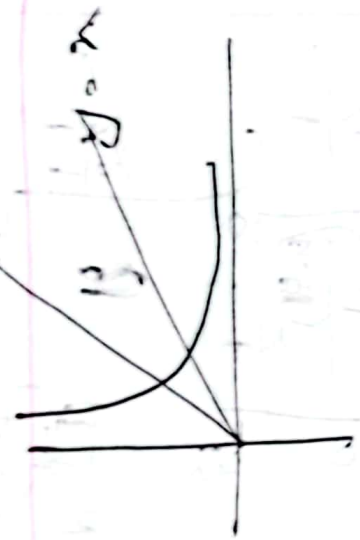
10

$$x = 2x$$

$$x - 2x = 0$$

$$x = 0$$

$$x = 0$$



$$\frac{1}{\sqrt{2}} \int_0^{2x} x^2 dy + \int_0^{1/\sqrt{2}} \frac{1}{x} x dy dx$$

$$\frac{1}{\sqrt{2}} \int_0^{2x} x^2 dy + \int_{1/\sqrt{2}}^1 x^2 \left[\frac{1}{y} \right]_x^1 dy$$

$$\frac{1}{\sqrt{2}} \int_0^{2x} x^2 dy + \int_{1/\sqrt{2}}^1 x^2 \left[\frac{1}{x} - \frac{1}{y} \right] dy$$

$$\frac{1}{\sqrt{2}} \int_0^{2x} x^2 dy + \int_{1/\sqrt{2}}^1 x^2 \left(\frac{1}{x} - \frac{1}{y} \right) dy$$

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$$\int_0^1 \frac{1}{\sqrt{x}} dx + \int_0^1 (x-x^3) dx$$

$$= \left[\frac{x^{1/2}}{1/2} \right]_0^1 + \left[\frac{x^2}{2} - \frac{x^4}{4} \right]_0^1$$

$$= \left[\frac{1}{\sqrt{x}} \right]_0^1 + \left[\frac{1}{2}x^2 - \frac{1}{4}x^4 \right]_0^1$$

$$= \left[\frac{1}{\sqrt{1}} - \frac{1}{\sqrt{0}} \right] + \left[\frac{1}{2} \left(1^2 - 0^2 \right) - \frac{1}{4} \left(1^4 - 0^4 \right) \right]$$

$$= \left[\frac{1}{1} - \frac{1}{0} \right] + \left[\frac{1}{2} - \frac{1}{4} \right]$$

$$= \frac{1}{1} + \frac{1}{4} - \frac{3}{16} = \frac{1+4-3}{16} = \frac{2}{16}$$

$$= \frac{1}{8} \text{ (Ans)}$$

$$\int_0^1 \frac{1}{\sqrt{x}} dx + \int_0^1 (x-x^3) dx$$

$$= \left[\frac{x^{1/2}}{1/2} \right]_0^1 + \left[\frac{x^2}{2} - \frac{x^4}{4} \right]_0^1$$

(12)

[14.3]

$$\boxed{28} \int_{-2}^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} e^{-(x^2+y^2)} dx dy$$

$$2 \int_0^{2\pi} \int_0^2 e^{-r^2} r dr d\theta$$

$$= \frac{1}{2} \int_0^{2\pi} \int_0^2 e^{-z} dz d\theta$$

$$= \frac{1}{2} \int_0^{2\pi} \left[\frac{e^{-z}}{-1} \right]_0^2 d\theta$$

$$= -\frac{1}{2} \int_0^{2\pi} (e^{-4} - e^0) d\theta$$

$$= -\frac{1}{2} \int_0^{2\pi} (e^{-4} - 1) d\theta$$

$$= -\frac{1}{2} \int_0^{2\pi} e^{-4} d\theta + \frac{1}{2} \int_0^{2\pi} d\theta$$

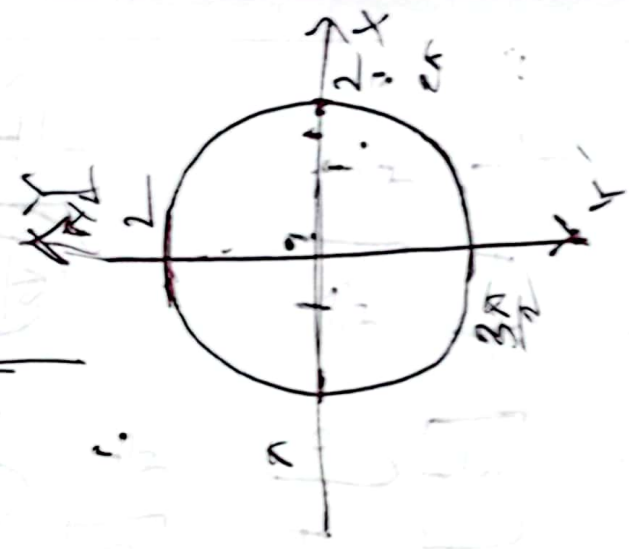
$$= -\frac{1}{2} e^{-4} [\theta]_0^{2\pi} + \frac{1}{2} [\theta]_0^{2\pi}$$

$$= -\frac{1}{2} e^{-4} (2\pi - 0) + \frac{1}{2} (2\pi - 0)$$

$$= -\frac{1}{2} e^{-4} (2\pi) + \pi$$

$$\begin{aligned} x &= \sqrt{4-y^2} \\ x^2 &= 4-y^2 \\ x^2+y^2 &= 4 \\ r^2 &= 4 \\ r &= \pm 2 \end{aligned}$$

$$\begin{aligned} y &= r \sin \theta \\ 2 &= 2 \sin \theta \\ \sin \theta &= 1 \end{aligned}$$



$$\begin{aligned} \theta &\in [0, 2\pi] \\ r &\in [0, 2] \end{aligned}$$

$$\begin{array}{l} r^2 = 2 \\ 2r dr = d2 \\ r dr = \frac{d2}{2} \\ \frac{r}{2} \Big|_0^2 \Big|_0^{2\pi} \end{array}$$

(19)

$$= \pi e^{4\pi} \rightarrow \pi (1 - e^4)$$

[29]

$$\int_0^2 \int_0^{\sqrt{2\pi-x^2}} \sqrt{x^2+y^2} \, dy \, dx$$

$$= \int_0^{\pi/2} \int_0^{2\cos\theta} \sqrt{r^2} \, r \, dr \, d\theta$$

$$= \int_0^{\pi/2} \int_0^{2\cos\theta} r^2 \, dr \, d\theta$$

$$= \int_0^{\pi/2} \left[\frac{r^3}{3} \right]_0^{2\cos\theta} d\theta$$

$$= \frac{1}{3} \int_0^{\pi/2} (2^3 \cos^3\theta - 0) d\theta$$

$$= \frac{1}{3} \int_0^{\pi/2} 8 \cos^3\theta \, d\theta$$

$$= \frac{8}{3} \int_0^{\pi/2} \cos^2\theta \cos\theta \, d\theta$$

$$= \frac{8}{3} \int_0^{\pi/2} (1 - \sin^2\theta) \cos\theta \, d\theta$$

upper limit of y

$$\sqrt{2\pi-x^2} = y$$

$$(\sqrt{2\pi-x^2})^2 = y^2$$

$$x^2 + y^2 = 2\pi$$

$$r^2 = 2\pi$$

$$r = \sqrt{2\pi}$$

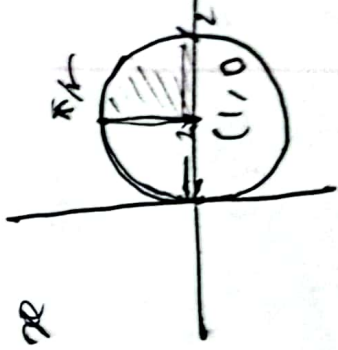
$$r^2 = 2\pi \cos\theta$$

$$r = 2\cos\theta$$

$$x^2 + y^2 - 2x = 0$$

$$x^2 - 2x + 1 + y^2 = 1$$

$$(x-1)^2 + y^2 = 1$$



(14)

$$\frac{8}{3} \int_0^1 (1-u^2) du$$

$$= \frac{8}{3} \left[du - \frac{2}{3} u^3 \right]_0^1$$

$$= \frac{8}{3} \left[u - \frac{2}{3} u^3 \right]_0^1$$

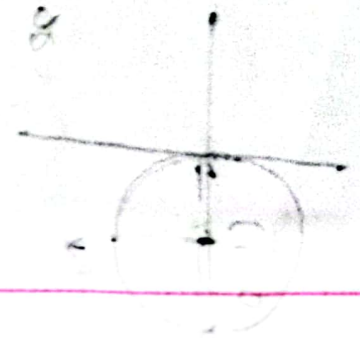
$$= \frac{8}{3} (1 - 0) - \frac{8}{9} (1 - 0)$$

$$= \frac{16}{9} \text{ (Ans)}$$

Let,

$$\sin \theta = u$$
$$\cos \theta d\theta = du$$

θ	$\frac{\pi}{2}$	0
u	1	0



[34]

$$\int_{-4}^0 \int_{-\sqrt{16-x^2}}^{\sqrt{16-x^2}} 3xy \, dy \, dx$$

Region is bounded by

$$x = -4$$

$$x = 0$$

$$x = r \cos \theta$$

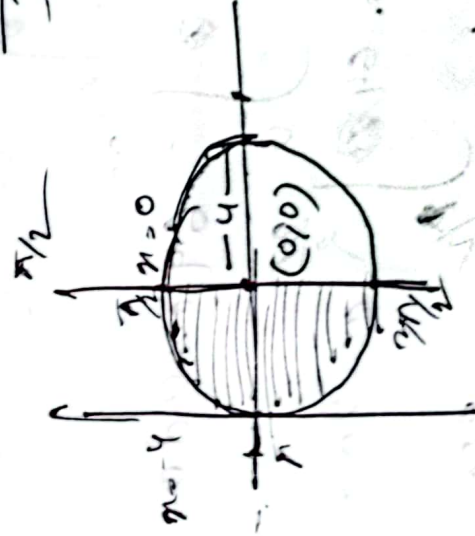
$$y = r \sin \theta$$

$$dx \, dy = r \, dr \, d\theta$$

Limits

$$r = 0 \rightarrow 4$$

$$\theta = \pi/2 \rightarrow \frac{3\pi}{2}$$



$x^2 + y^2 = 4^2$
center (0,0)
radius 4

(15)

$$y = \pm \sqrt{16-x^2}$$

$$y^2 = 16 - x^2$$

$$x^2 + y^2 = 16 \text{ (circle)}$$

$$r^2 = 16$$

$$r = \pm 4$$

$$x = r \cos \theta$$

$$0 = 4 \cos \theta$$

$$\theta = \pi/2$$

Again,

$$-4 = 4 \cos \theta$$

$$\cos \theta = -1$$

$$\theta = \pi/2 + \pi$$

$$= \frac{3\pi}{2}$$

(12)

Now $\int_{-\pi/2}^{\pi/2} \frac{3}{2} \cos \theta \, d\theta$

$= \frac{3}{2} \int_{-\pi/2}^{\pi/2} \cos \theta \, d\theta$

$= \frac{3}{2} \left[\sin \theta \right]_{-\pi/2}^{\pi/2}$

$= \frac{3}{2} \left[\sin \frac{\pi}{2} - \sin \left(-\frac{\pi}{2}\right) \right]$

$= \frac{3}{2} \left[1 - (-1) \right]$

$= \frac{3}{2} \left[1 + 1 \right]$

$= \frac{3}{2} \times 2$

$= 3$

$\therefore \int_{-\pi/2}^{\pi/2} \frac{3}{2} \cos \theta \, d\theta = 3$

$\therefore \int_{-\pi/2}^{\pi/2} \frac{3}{2} \cos \theta \, d\theta = 3$

(17)

14.5

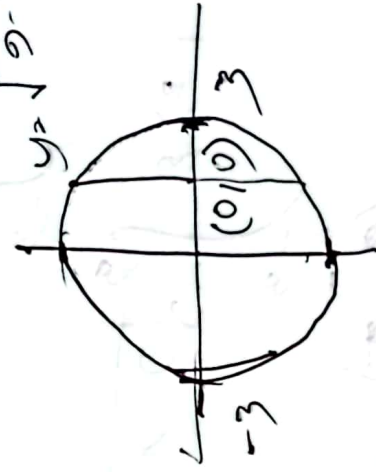
Example 3:-

Use a triple integral to find the volume of the solid within the cylinder $x^2 + y^2 = 9$ and between the planes $z = 1$ and $x + z = 5$.

Solution:-

The solid is

$$\int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_1^{5-x} dz \, dy \, dx$$



$$\int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} [z]_1^{5-x} dy \, dx$$

$$\int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} (5-x-1) dy \, dx$$

$$\int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} (x-4) dy \, dx$$

$$x^2 + y^2 = 9$$

$$y^2 = 9 - x^2$$

$$y = \pm \sqrt{9 - x^2}$$

$$x^2 + y^2 = 3^2$$

$$r = \pm 3$$

$$x + z = 5$$

$$z = 5 - x$$

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$$= \int_{-3}^3 \sqrt{9-x^2} (4-x) dx$$

$$= \int_{-3}^3 (8-2x) \sqrt{9-x^2} dx$$

$$= 8 \int_{-3}^3 \sqrt{9-x^2} dx - \int_{-3}^3 2x \sqrt{9-x^2} dx$$

$$= 8 \left(\frac{9}{2} \pi \right) - \int_{-3}^3 2x \sqrt{9-x^2} dx$$

$$= 8 \left(\frac{9}{2} \pi \right) - 0 = 36\pi$$

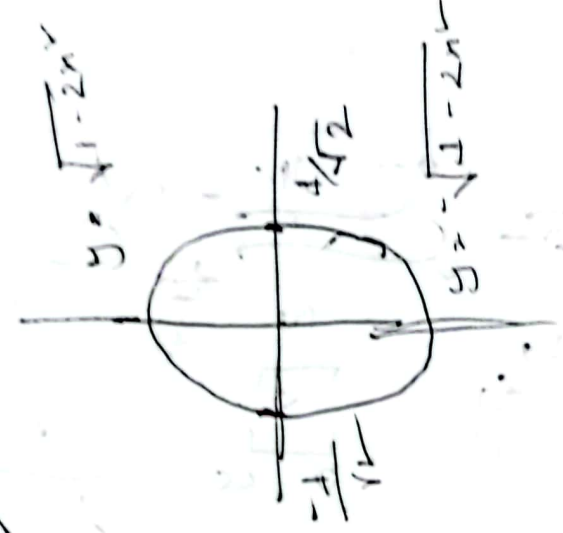
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Ex-4) $z = 5x^2 + 5y^2 \quad z = 6 - 7x^2 - y^2$

Solution:-

$$5x^2 + 5y^2 = 6 - 7x^2 - y^2$$

$$2x^2 + y^2 = 1$$



$$= \int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} \int_{-\sqrt{1-2x^2}}^{\sqrt{1-2x^2}} (6 - 7x^2 - y^2) dy dx$$

$$= \int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} \int_{-\sqrt{1-2x^2}}^{\sqrt{1-2x^2}} (6 - 12x^2 - 6y^2) dy dx$$

$$= \int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} \left[(1 - 2x^2)y - 2y^3 \right]_{-\sqrt{1-2x^2}}^{\sqrt{1-2x^2}} dx$$

$$= \frac{1}{8} \int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} (1 - 2x^2)^{3/2} dx = \frac{8}{\sqrt{2}} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^4 \theta d\theta$$

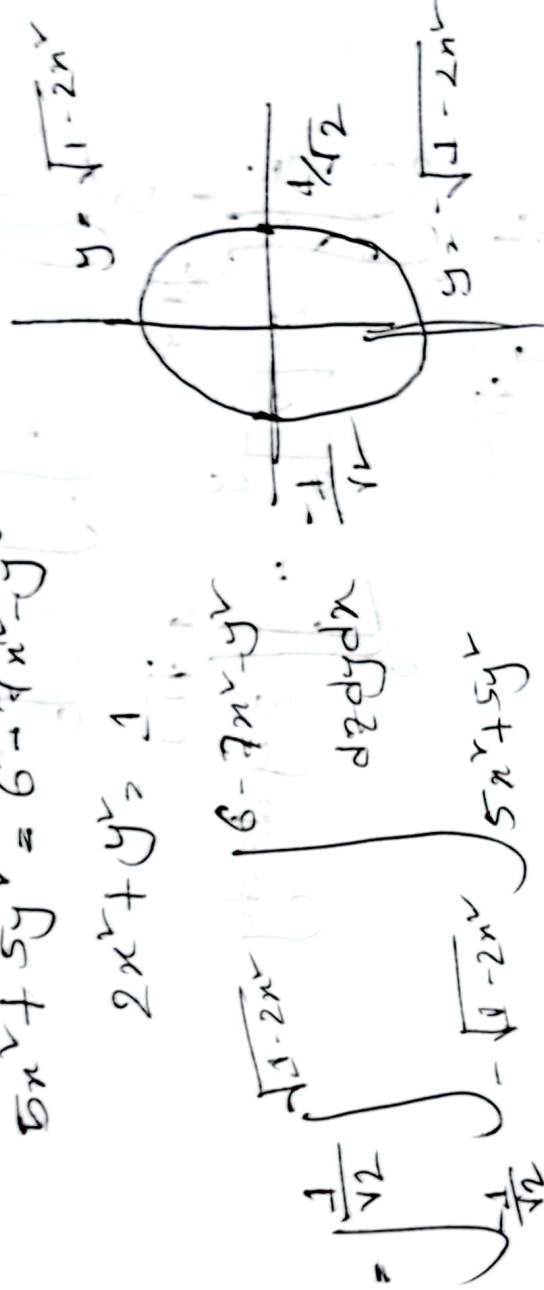
(19)

[Ex-4] $z = 5x^2 + 5y^2 \quad z = 6 - 7x^2 - y^2$

Solution:-

$$5x^2 + 5y^2 = 6 - 7x^2 - y^2$$

$$2x^2 + y^2 = 1$$



$$= \int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} \int_{-\sqrt{1-2x^2}}^{\sqrt{1-2x^2}} (6 - 7x^2 - y^2) dy dx$$

$$= \int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} \int_{-\sqrt{1-2x^2}}^{\sqrt{1-2x^2}} (6 - 12x^2 - 6y^2) dy dx$$

$$= \int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} 6 \left[(1 - 2x^2)y - 2y^3 \right]_{-\sqrt{1-2x^2}}^{\sqrt{1-2x^2}} dx$$

$$= 8 \int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} (1 - 2x^2)^{3/2} dx = \frac{8}{\sqrt{2}} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^4 \theta d\theta$$

$$= \frac{8}{\sqrt{2}}$$

Example-5

Solution:- The solid is bounded in the back by the plane $x=0$ and in the front by the plane

$$x=y$$

$$\iiint_V z \, dv = \int_0^1 \int_0^y \int_0^{\sqrt{1-y}} z \, dx \, dz \, dy$$

$$= \int_0^1 \left[x \right]_0^y \cdot \left[\frac{z^2}{2} \right]_0^{\sqrt{1-y}} dy = \int_0^1 z y \, dz \, dy$$

$$= \int_0^1 \left[\frac{1}{2} z^2 y \right]_0^{\sqrt{1-y}} dy$$

$$= \int_0^1 y \cdot \frac{(1-y)}{2} dy$$

$$= \frac{1}{2} \int_0^1 y - y^2 dy$$

$$= \frac{1}{2} \left[\frac{y^2}{2} - \frac{y^3}{3} \right]_0^1 = \frac{1}{2} \left(\frac{1^2}{2} - \frac{1^3}{3} \right)$$

$$= \frac{1}{2} \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}$$

$$= \frac{2-1}{8} = \frac{1}{8}$$

Exercise

$$\boxed{a} \int_{-1/3}^{1/3} \int_{-1/2}^{1/2} \int_0^{\pi} 2x \sin xy \, dz \, dy \, dx$$

$$= \int_{-1/3}^{1/3} \int_{-1/2}^{1/2} x \sin xy \left[\frac{z^2}{2} \right]_0^{\pi} dy \, dx$$

$$= \int_{-1/3}^{1/3} \int_{-1/2}^{1/2} x \sin xy \left(\frac{1}{2} - 0 \right) dy \, dx = \int_{-1/3}^{1/3} \int_{-1/2}^{1/2} \frac{1}{2} x \sin xy \, dy \, dx$$

$$= \int_{-1/3}^{1/3} \left[\frac{\cos xy}{x} \right]_{-1/2}^{1/2} dx = - \int_{-1/3}^{1/3} [\cos xy]_0^{\pi} dx$$

$$= - \int_{-1/3}^{1/3} (\cos \pi x - \cos x \cdot 0) dx$$

$$= - \int_{-1/3}^{1/3} (\cos \pi x - 1) dx = - \int_{-1/3}^{1/3} \cos \pi x \, dx + \int_{-1/3}^{1/3} 1 \, dx$$

$$= - \left[\frac{\sin \pi x}{\pi} \right]_{-1/3}^{1/3} + \left[x \right]_{-1/3}^{1/3}$$

$$= \frac{1}{2} \left(\frac{1}{2} - \frac{1}{12} \right) + \frac{1}{\pi} \left(\frac{\sin \frac{\pi}{2}}{2} - \frac{\sin \frac{\pi}{3}}{3} \right)$$

$$= \frac{1}{12} + \frac{1}{2} \left(\frac{1 - \sqrt{3}/2}{\pi} \right)$$

$$= \frac{1}{12} + \frac{1}{2} \left(\frac{2 - \sqrt{3}}{2\pi} \right) = \frac{1}{12} + \frac{\sqrt{3} - 2}{4\pi} \quad (\text{Ans})$$

[3]

$$= \int_0^2 \int_{-1}^1 \int_{-1}^1 yz \, dx \, dz \, dy$$

$$= \int_0^2 \int_{-1}^1 yz \left[x \right]_{-1}^1 dz \, dy$$

$$= \int_0^2 \int_{-1}^1 yz [x^2]_{-1}^1 dz \, dy$$

$$= \int_0^2 \int_{-1}^1 yz (2 + 1) dz \, dy$$

$$= \int_0^2 \int_{-1}^1 (yz^2 + yz) dz \, dy$$

$$= \int_0^2 \int_{-1}^1 y(z^2 + z) dz \, dy$$

$$= \int_0^2 \left[yz^2 + yz^2 + yz^2 + yz^2 \right]_{-1}^1 dy$$

$$= \int_0^2 y \left[\frac{z^3}{3} + \frac{z^3}{3} + \frac{z^3}{3} + \frac{z^3}{3} \right]_{-1}^1 dy$$

$$= \int_0^2 y \left(\frac{y^3}{3} + \frac{y^3}{3} + \frac{y^3}{3} + \frac{y^3}{3} \right) dy$$

$$= \int_0^2 \frac{y^4}{3} dy + \int_0^2 \frac{y^4}{3} dy$$

$$= \left[\frac{y^5}{15} + \frac{y^5}{15} \right]_0^2$$

$$= \frac{1}{3} \left[\frac{y^8}{8} \right]_0^2 + \frac{1}{2} \left[\frac{y^5}{5} \right]_0^2 - \frac{1}{6} [y]_0^2$$
~~$$= \frac{1}{3} \left(\frac{2^8}{8} - \frac{0}{8} \right) + \frac{1}{2} \left(\frac{2^5}{5} - \frac{0}{5} \right) - \frac{1}{6} (2 - 0)$$

$$= \frac{1}{3} (2^8 - 0) + \frac{1}{2} (2^5 - 0) - \frac{1}{6} (2 - 0)$$

$$= \frac{1}{3} (256 - 0) + \frac{1}{2} (32 - 0) - \frac{1}{6} (2 - 0)$$~~

$$= \frac{1}{3} \left[\frac{y^8}{8} \right]_0^2 + \frac{1}{2} [y]_0^2 - \frac{1}{6} [y]_0^2$$

$$= \frac{1}{3} \left(\frac{256}{8} \right) + \frac{1}{2} \left(\frac{64}{6} \right) - \frac{1}{6} (2)$$

$$= \frac{32}{3} + \frac{16}{3} - \frac{1}{3} = \frac{47}{3}$$

18

$$\int_1^2 \int_0^{\log y} \frac{y}{x^2 + y} dx dy$$

$$\int_1^2 \int_0^{\log y} \frac{1}{x^2 + y} dx dy$$

$$\int_1^2 \left[\frac{1}{y} \tan^{-1} \frac{x}{y} \right]_0^{\log y} dy$$

$$= \int_1^2 \int_1^2 (\tan^{-1} y - \tan^{-1} 0) dy dz$$

$$= \int_1^2 \left\{ \int_1^2 \frac{\pi}{3} dy - \int_1^2 0 dy \right\} dz$$

$$= \int_1^2 \left[\frac{\pi}{3} y \right]_1^2 dz$$

$$= \frac{\pi}{3} \int_1^2 (2-1) dz = \frac{\pi}{3} [2]_1^2 = \frac{\pi}{3} \left[\frac{2^2}{2} \right]_1^2$$

$$= \frac{2\pi}{3} - \frac{\pi}{6} (4-1)$$

$$= \frac{2\pi}{3} - \frac{\pi \times 3}{6} = \frac{2\pi}{3} - \frac{\pi}{2} = \frac{\pi}{6}$$

steps

10 $\iiint y \, dv$, where G is the solid in the first octant enclosed by the plane $z=y$, the xy -plane, and the parabolic cylinder $y=1-x^2$.

$$\iiint_G y \, dv$$

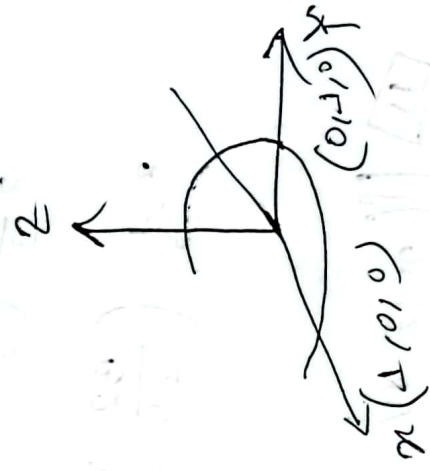
$$= \int_{-1}^1 \int_0^{1-x^2} \int_0^y y \, dz \, dy \, dx$$

$$= \int_{-1}^1 \int_0^{1-x^2} y [z]_0^y \, dy \, dx$$

$$= \int_{-1}^1 \int_0^{1-x^2} y^2 \, dy \, dx$$

$$= \int_{-1}^1 \left[\frac{y^3}{3} \right]_0^{1-x^2} dx = \frac{1}{3} \int_{-1}^1 (1-x^2)^3 dx$$

$$= \frac{1}{3} \int_{-1}^1 (-x^6 + 3x^4 - 3x^2 + 1) dx$$



$$= \frac{1}{3} \left[\left(-\frac{1}{7} + \frac{3}{5} - 1 + 1 \right) - \left(\frac{1}{7} - \frac{3}{5} + 1 - 1 \right) \right]$$

$$= \frac{1}{3} \left(-\frac{1}{7} + \frac{3}{5} - \frac{1}{7} + \frac{3}{5} \right)$$

$$= \frac{1}{3} \left(\frac{6}{5} - \frac{2}{7} \right) = \frac{1}{3} \left(\frac{42-10}{35} \right)$$

$$= \frac{1}{3} \left(\frac{32}{35} \right) = \frac{32}{105}$$

$$\boxed{11} \iiint_G xyz \, dV$$

$$z = 2 - x^2$$

$$z = 0$$

$$y = x$$

$$y = 0$$

$$z = 2 - x^2$$

$$z = 0$$

$$y = x$$

$$y = 0$$

$$2 - x^2 = 0 \Rightarrow x^2 = 2$$

$$x = \sqrt{2}$$

x varies from 0 to $\sqrt{2}$, the region is in the first quadrant.

$$\int_0^{\sqrt{2}} \int_0^x \int_0^{2-x} xyz \, dz \, dy \, dx$$

$$\int_0^{\sqrt{2}} \int_0^x \left[xy \left[\frac{z^2}{2} \right]_0^{2-x} \right] dy \, dx$$

$$\int_0^{\sqrt{2}} \int_0^x (2-x^2) xy \, dy \, dx$$

$$\int_0^{\sqrt{2}} \int_0^x (2-x^2) x y \, dy \, dx = \int_0^{\sqrt{2}} \left(4 - 4x^2 + x^4 \right) x y \, dy \, dx$$

$$\int_0^{\sqrt{2}} \int_0^x (2x - x^3) y \, dy \, dx$$

$$\int_0^{\sqrt{2}} (2x - x^3) \left[\frac{y^2}{2} \right]_0^x dx$$

$$\int_0^{\sqrt{2}} (2x - x^3) x^2 dx$$

$$\int_0^{\sqrt{2}} \frac{2x^3 - x^5}{2} dx$$

$$= \int_0^{\sqrt{2}} (4 - 4x^2 + x^4) \int_0^x y dy dx$$

$$= \int_0^{\sqrt{2}} (4x - 4x^3 + x^5) \left[\frac{y^2}{2} \right]_0^x dx$$

$$\Rightarrow \frac{1}{2} \int_0^{\sqrt{2}} (4x - 4x^3 + x^5) dx$$

$$= \frac{1}{2} \int_0^{\sqrt{2}} (4x^3 - 4x^4 + x^5) dx$$

$$= \frac{1}{2} \left(4x^3 - \frac{4}{5}x^5 + \frac{1}{6}x^6 \right) \Big|_0^{\sqrt{2}}$$

$$= \frac{1}{2} \left[\frac{4}{3}x^3 - \frac{4}{5}x^5 + \frac{1}{6}x^6 \right]_0^{\sqrt{2}}$$

$$= \frac{1}{2} \left(\frac{4}{3}(\sqrt{2})^3 - \frac{4}{5}(\sqrt{2})^5 + \frac{1}{6}(\sqrt{2})^6 \right)$$

$$= \frac{1}{2} \left(\frac{4}{3}(2\sqrt{2}) - \frac{4}{5}(4\sqrt{2}) + \frac{1}{6}(4) \right)$$

$$= \frac{1}{2} \left(\frac{8\sqrt{2}}{3} - \frac{16\sqrt{2}}{5} + \frac{2}{3} \right)$$

$$\Rightarrow \frac{1}{2} \left(\frac{8\sqrt{2}}{3} - \frac{16\sqrt{2}}{5} + \frac{2}{3} \right)$$

[12]

$$\iiint_G \cos(z/y) dv \rightarrow G = \left. \begin{array}{l} \frac{\pi}{6} \leq y \leq \frac{\pi}{2} \\ y \leq x < \frac{\pi}{2} \\ 0 \leq z \leq xy \end{array} \right\}$$

⇒ From the Fubini's

Theorem we can say that

if G is rectangular box defined by the

inequalities ~~as~~

$$\iiint_G \cos(z/y) dv = \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \int_0^{xy} \int_0^{xy} \cos(z/y) dz dx dy$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \int_0^{xy} \left[\sin \frac{z}{y} \right]_0^{xy} dy dx$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \int_0^{xy} \sin(xy) dx dy$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \left[-\cos(xy) \right]_0^{xy} dy = \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (-\cos(xy) + 1) dy$$

$$\int u^v = \frac{u^{v+1}}{v+1}$$

$$= - \int y [\cos x]_{\pi/6}^{\pi/2} dy$$

$$= - \int_{\pi/6}^{\pi/2} y (\cos \frac{\pi}{2} - \cos y) dy$$

$$= - \int_{\pi/6}^{\pi/2} y [\sin y - \frac{1}{2}] dy$$

$$= \int_{\pi/6}^{\pi/2} y \sin y - \frac{y}{2} dy$$

$$= \left[-y \cos y + \sin y - \frac{y^2}{4} \right]_{\pi/6}^{\pi/2}$$

$$= \left[-\frac{\pi}{2} \cos \frac{\pi}{2} + \sin \frac{\pi}{2} - \frac{(\frac{\pi}{2})^2}{4} \right] - \left[-\frac{\pi}{6} \cos \frac{\pi}{6} + \sin \frac{\pi}{6} - \frac{(\frac{\pi}{6})^2}{4} \right]$$

$$= \frac{5\pi - \cancel{12\pi}}{\cancel{12}} - \frac{\sqrt{3}}{2} \Rightarrow \frac{5\pi}{12} - \frac{\sqrt{3}}{2}$$

$$\Rightarrow \frac{5\pi - 6\sqrt{3}}{12}$$

14.6

Example 1

$$\iiint dV$$

$$V = \int_0^{2\pi} \int_0^3 \int_0^{\sqrt{25-r^2}} r \, dz \, dr \, d\theta$$

$$2 \cdot \int_0^{2\pi} \int_0^3 r [z]_0^{\sqrt{25-r^2}} \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^3 r \sqrt{25-r^2} \, dr \, d\theta$$

$$= \int_0^{2\pi} \left[-\frac{1}{3} (25-r^2)^{3/2} \right]_{r=0}^3 \, d\theta$$

$$= \int_0^{2\pi} \left(-\frac{1}{3} (25-9)^{3/2} + \frac{1}{3} (25)^{3/2} \right) \, d\theta = \int_0^{2\pi} \frac{64}{3} \, d\theta$$

$$= \frac{128}{3} \pi$$

Example - 91-

$$\int_{-3}^3 \int_{-\sqrt{9-r^2}}^{\sqrt{9-r^2}} \int_{-\sqrt{9-r^2}}^{\sqrt{9-r^2}} x^2 dz dy dx$$

$$\int_{-3}^3 \int_{-\sqrt{9-r^2}}^{\sqrt{9-r^2}} \int_{-\sqrt{9-r^2}}^{\sqrt{9-r^2}} r^2 \cos^2 \theta r dz dr d\theta$$

$$\int_{-3}^3 \int_{-\sqrt{9-r^2}}^{\sqrt{9-r^2}} \int_{-\sqrt{9-r^2}}^{\sqrt{9-r^2}} r^3 \cos^2 \theta dz dr d\theta$$

$$\int_{-3}^3 \int_{-\sqrt{9-r^2}}^{\sqrt{9-r^2}} \left[\frac{r^4 \cos^2 \theta}{2} \right]_{-\sqrt{9-r^2}}^{\sqrt{9-r^2}} dr d\theta$$

$$\int_{-3}^3 \int_{-\sqrt{9-r^2}}^{\sqrt{9-r^2}} (9r^3 - r^5) \cos^2 \theta dr d\theta$$

$$\int_{-3}^3 \left[\frac{9r^4}{4} - \frac{r^6}{6} \right]_{-\sqrt{9-r^2}}^{\sqrt{9-r^2}} \cos^2 \theta d\theta$$

$$\int_{-3}^3 \left[\frac{9r^4}{4} - \frac{r^6}{6} \right]_{r=0}^{\sqrt{9-r^2}} \cos^2 \theta d\theta$$

$$= \frac{243}{4} \int_0^\pi \cos^2 \theta d\theta$$

$$= \frac{243}{4} \int_0^\pi \frac{1}{2} (1 + \cos 2\theta) d\theta$$

$$= \frac{243\pi}{4}$$

Ex-4

$$\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2-2}}^{\sqrt{4-x^2-y^2}} dz dy dx$$

$$= \int_0^{2\pi} \int_0^2 \int_0^5 \cos \phi \sin \phi d\phi dr d\theta$$

$$= \int_0^{2\pi} \int_0^{\frac{2\sqrt{2}}{3}} \cos \phi \sin \phi d\phi dr d\theta$$

$$= \frac{2\sqrt{2}}{3} \int_0^{2\pi} \left[-\frac{1}{3} \cos^3 \phi \right]_0^{\pi} d\theta$$

$$= \frac{2\sqrt{2}}{9} \int_0^{2\pi} d\theta = \frac{64\pi}{9}$$

$$\frac{64\pi}{9}$$

Exercise - 10

10) $x^2 + y^2 + z^2 = 1$ — (1)

and below by the cone $z = \sqrt{x^2 + y^2}$

$$z = \sqrt{x^2 + y^2}$$

$$z^2 = x^2 + y^2$$

from (1) $\Rightarrow$

$$z^2 + z^2 = 1$$

$$z^2 = \frac{1}{2}$$

$$\therefore z = \frac{1}{\sqrt{2}}$$

$$\therefore z = \sqrt{x^2 + y^2}$$

$$\left(\frac{1}{\sqrt{2}}\right)^2 = \sqrt{x^2 + y^2}$$

$$r^2 = \frac{1}{2}$$

$$r = \frac{1}{\sqrt{2}}$$

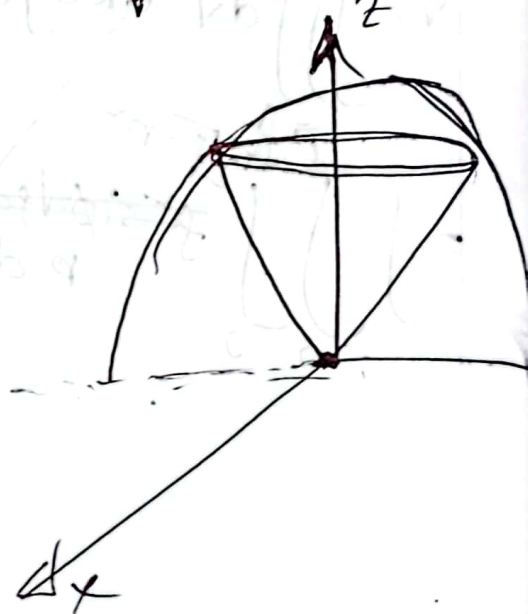
Again,

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$0 \leq r < \frac{1}{\sqrt{2}}$$

$$0 \leq \theta \leq 2\pi$$



$$\sqrt{x^2+y^2} \leq z \leq \sqrt{1-x^2-y^2}$$

$$r \leq z \leq \sqrt{1-r^2}$$

$$\text{Volume} = \int_0^{2\pi} \int_0^{\frac{1}{\sqrt{2}}} \int_r^{\sqrt{1-r^2}} r \, dz \, dr \, d\theta$$

$$\Rightarrow \int_0^{2\pi} \int_0^{\frac{1}{\sqrt{2}}} r \left[z \right]_r^{\sqrt{1-r^2}} dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^{\frac{1}{\sqrt{2}}} r (\sqrt{1-r^2} - r) dr \, d\theta$$

$$= 2 \int_0^{2\pi} \left[\int_0^{\frac{1}{\sqrt{2}}} (r(\sqrt{1-r^2}) - r^2) dr \right] d\theta$$

$$= 2 \int_0^{2\pi} \left[\frac{r^2}{2} \cdot \frac{2}{3} (1-r^2)^{\frac{3}{2}} \right]_0^{\frac{1}{\sqrt{2}}} - \left[\frac{r^3}{3} \right]_0^{\frac{1}{\sqrt{2}}} d\theta$$

$$= 2 \int_0^{2\pi} \left[\frac{r^2}{3} \right]_{\frac{1}{\sqrt{2}}} \left(1 - \frac{1}{\sqrt{2}} \right)^{\frac{3}{2}} - \left[\frac{\left(\frac{1}{\sqrt{2}}\right)^3}{3} \right] d\theta$$

$$= 2 \int_0^{2\pi} \left[\frac{1}{2} \cdot \frac{1}{3} (1-r^2)^{\frac{3}{2}} - \frac{1}{6\sqrt{2}} \right] d\theta$$

$$\int_0^{2\pi} \left[\frac{1}{3} \left(\frac{1}{2\sqrt{2}} - 1 \right) - \left(\frac{1}{6\sqrt{2}} \right) \right]$$

$$\Rightarrow \frac{1}{3} - \frac{1}{3\sqrt{2}} \int_0^{2\pi} d\theta$$

$$\Rightarrow \left(\frac{2\sqrt{2} - 2}{3\sqrt{2}} \right) \pi$$

$$\text{or } (1 - \sqrt{2}) \pi$$

$$\text{or } (1 - \sqrt{2}) \pi$$

$$\Rightarrow \left[\frac{1}{3} \left(\frac{1}{2\sqrt{2}} - 1 \right) - \left(\frac{1}{6\sqrt{2}} \right) \right] \int_0^{2\pi} d\theta$$

$$\Rightarrow \left[\frac{1}{3} - \frac{1}{3\sqrt{2}} \right] \int_0^{2\pi} d\theta$$

14

given that $\phi = \frac{\pi}{4}$ $\rho = 1$ and $\rho = 2$.

Using spherical co-ordinates

$$0 \leq \phi \leq \frac{\pi}{4}, \quad 0 \leq \theta \leq 2\pi, \quad 1 \leq \rho \leq 2$$

$$\text{Volume} \rightarrow \int_0^{2\pi} \int_0^{\frac{\pi}{4}} \int_1^2 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

$$\rightarrow \int_0^{2\pi} \int_0^{\frac{\pi}{4}} \left[\frac{\rho^3}{3} \right]_1^2 \sin \phi \, d\phi \, d\theta$$

$$\rightarrow \int_0^{2\pi} \int_0^{\frac{\pi}{4}} \left(\frac{2^3}{3} - \frac{1}{3} \right) \sin \phi \, d\phi \, d\theta$$

$$\rightarrow \int_0^{2\pi} \left(\frac{8}{3} - \frac{1}{3} \right) \sin \phi \, d\phi \, d\theta$$

$$\rightarrow -\frac{7}{3} \int_0^{2\pi} [\cos \phi]_0^{\frac{\pi}{4}} \, d\theta$$

$$\rightarrow -\frac{7}{3} \int_0^{2\pi} \left(\cos \frac{\pi}{4} - \cos 0 \right) \, d\theta$$

$$\rightarrow -\frac{7}{3} \int_0^{2\pi} \left(\frac{1}{\sqrt{2}} - 1 \right) \, d\theta \quad \rightarrow -\frac{7}{3} \left(\frac{1}{\sqrt{2}} - 1 \right) (2\pi - 0)$$

$$\rightarrow -\frac{7\pi}{3} \left(\frac{1}{\sqrt{2}} - 1 \right) = \frac{7\pi}{3} (2 - \frac{1}{\sqrt{2}})$$

17

$$\int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{\sqrt{a^2-x^2-y^2}} x^2 dz dy dx$$

$$= \int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{\sqrt{a^2-x^2-y^2}} x^2 dz dy dx$$

Here -

$$0 \leq x \leq \sqrt{a^2 - y^2 - z^2}$$

$$0 \leq y \leq \sqrt{a^2 - x^2}$$

$$0 \leq z \leq a$$

$$S = z = \sqrt{a^2 - x^2 - y^2}$$

xy plane $z = 0$

if we change the co-ordinates into the cylindrical co-ordinates as -

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = \sqrt{a^2 - x^2 - y^2}$$

$$= \sqrt{a^2 - (r^2 \cos^2 \theta + r^2 \sin^2 \theta)}$$

$$= \sqrt{a^2 - r^2}$$

$$= \sqrt{a^2 - r^2}$$

both xy plane intersect at -

$$0 \leq a^2 - r^2$$

$$r^2 \leq a^2$$

$$r \leq a$$

the region R is bounded in the first octant
then the Region R can be expressed as

$$0 \leq z \leq a^2 - r^2$$

$$0 \leq r \leq a$$

$$0 \leq \theta \leq \pi/2$$

Now

$$\int_0^a \int_0^{\sqrt{a^2-r^2}} \int_0^{a^2-r^2-y^2} x^2 dz dy dx$$

$$= \int_0^a \int_0^{\sqrt{a^2-r^2}} \int_0^{a^2-r^2} r^2 \cos^2 \theta r dz dr d\theta$$

$$= \int_0^{\pi/2} \int_0^a \int_0^{a^2-r^2} r^3 \cos^2 \theta dz dr d\theta$$

$$= \int_0^{\pi/2} \int_0^a r^3 \cos^2 \theta [z]_0^{a^2-r^2} dr d\theta$$

$$= \int_0^{\pi/2} \int_0^a (b^3 \cos^2 \theta) (a^2 - r^2) 2r dr d\theta$$

$$= \int_0^{\pi/2} \int_0^a \cos^2 \theta (a^2 r - r^3) dr d\theta$$

$$= \int_0^{\pi/2} \cos^2 \theta \left[a^2 \left[\frac{r^2}{2} \right] - \frac{r^4}{4} \right]_0^a d\theta$$

$$= \int_0^{\pi/2} \cos^2 \theta \left(\frac{a^4}{2} - \frac{a^4}{4} \right) d\theta$$

$$= \int_0^{\pi/2} \cos^2 \theta \left(\frac{a^4}{4} \right) d\theta$$

$$= \frac{a^4}{4} \int_0^{\pi/2} (\cos^2 \theta) d\theta$$

$$= \frac{a^4}{4} \int_0^{\pi/2} \frac{(\cos(2\theta) + 1)}{2} d\theta$$

$$= \frac{a^4}{8} \int_0^{\pi/2} (\cos(2\theta) + 1) d\theta$$

$$= \frac{a^6}{24} \left[\frac{1}{2} \sin 2\theta + 0 \right]_0^{\frac{\pi}{2}}$$

$$= \frac{a^6}{24} \left[\frac{1}{2} \sin \left(2 \frac{\pi}{2} \right) - \sin(2 \times 0) + \frac{\pi}{2} - 0 \right]$$

$$= \frac{a^6}{24} \left[\frac{\sin \pi}{2} - \sin(0) + \frac{\pi}{2} - 0 \right]$$

$$= \frac{a^6}{24} \left[\frac{1}{2} (0 - 0) + \frac{\pi}{2} - 0 \right]$$

$$= \frac{a^6}{24} \left[\frac{\pi}{2} \right]$$

$$= \frac{a^6 \pi}{48}$$

20. $\int_{-3}^3 \int_{-\sqrt{9-y^2}}^{\sqrt{9-y^2}} \int_{-\sqrt{9-x^2-y^2}}^{\sqrt{9-x^2-y^2}} \sqrt{x^2+y^2+z^2} dz dx dy$

this is the equation of full circle -

$$x^2 + y^2 + z^2 = 9$$

$$dx dy dz = \rho^2 \sin \phi d\rho d\phi d\theta$$

$$z = \sqrt{9 - x^2 - y^2}$$

$$z^2 = 9 - x^2 - y^2$$

$$x^2 + y^2 + z^2 = 9$$

$$\rho^2 = 9$$

$$\rho = 3$$

$$x = \sqrt{9 - y^2}$$

$$x^2 = 9 - y^2$$

$$x^2 + y^2 = 9$$

$$\rho = 3$$

$$0 \leq y \leq 3$$

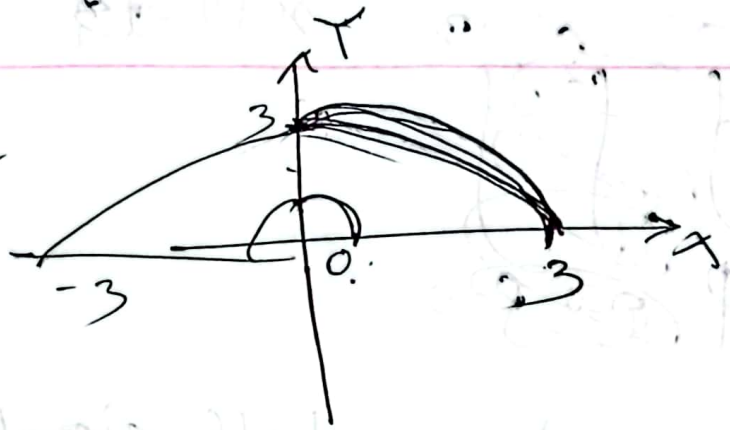
$$0 \leq x \leq \sqrt{9-y^2}$$

$$x = \sqrt{9-y^2}$$

$$x^2 = 9-y^2$$

$$x^2 + y^2 = 9 \quad \text{--- (1)}$$

$$r = 3$$



So it is in positive x axis so ~~0 to 2~~

$$0 \leq \theta \leq \frac{\pi}{4}$$

$\sqrt{x^2+y^2} \leq r \leq \sqrt{9-x^2-y^2}$
$0 \leq \theta \leq \frac{\pi}{4}$
$0 \leq \phi \leq \frac{\pi}{4}$

$$\begin{aligned} \sqrt{9-y^2} &= x \\ x^2 + y^2 &= 9 \\ r^2 &= 9 - x^2 - y^2 \\ r^2 &= 9 - 9 \\ r &= 0 \end{aligned}$$

$$\sqrt{x^2+y^2} = 2$$

$$x^2 + y^2 = 2^2$$

$$x^2 + y^2 + z^2 = 2^2$$

$$x^2 + y^2 + z^2 = 2^2$$

$$r = \sqrt{2}$$

$$r = \sqrt{2}$$

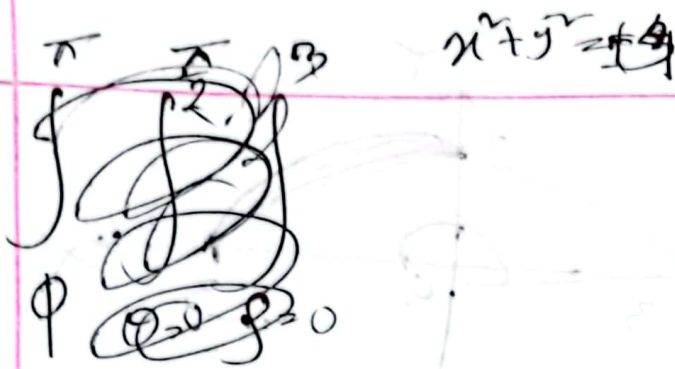
$$\cos \phi = \frac{1}{\sqrt{2}}$$

$$\phi = \frac{\pi}{4}$$

$$\phi = \frac{\pi}{4}$$

$$z^2 = 9 - (x^2 + y^2)$$

$$z^2 = 9 - 9 \quad (\text{from eq (1)})$$



This is a half circle. Therefore

$$\int_0^{2\pi} \int_0^{\pi} \int_0^3 \rho^3 \sin \phi \, d\rho \, d\phi \, d\theta$$

$$2) \int_0^{2\pi} \int_0^{\pi} \left[\frac{\rho^4}{4} \right]_0^3 \sin \phi \, d\phi \, d\theta$$

$$= \frac{1}{4} \int_0^{2\pi} \int_0^{\pi} (81 - 0) \sin \phi \, d\phi \, d\theta$$

$$= \frac{81}{4} \int_0^{2\pi} \left[-\cos \phi \right]_0^{\pi} d\theta$$

$$= \frac{81}{4} \int_0^{2\pi} (\cos \pi - \cos 0) d\theta$$

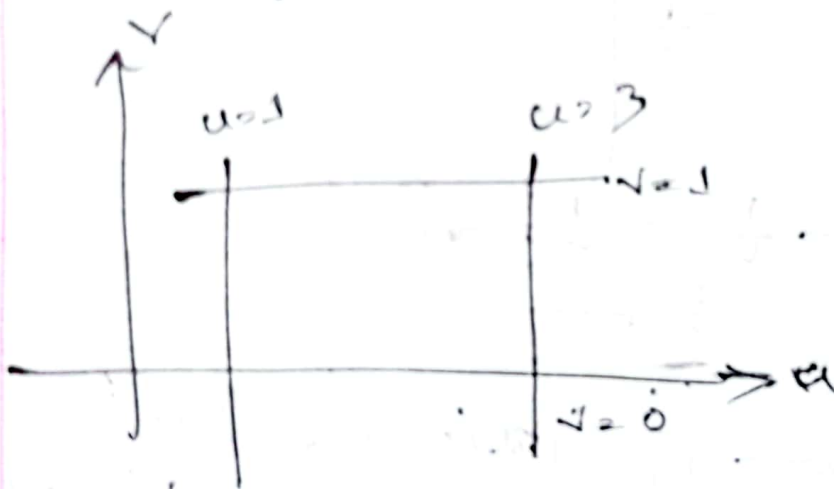
$$= \frac{-81}{4} \int_0^{2\pi} (-1 - 1) d\theta$$

$$= \frac{81}{4} \times 2 \times [2\pi - 0] = 81\pi$$

Example 2:-

$$\oint \frac{x-y}{x+y} dA$$

The diagram is now



$$u = x+y$$

$$v = x-y$$

$$x-y=0$$

$$x-y=1$$

$$x+y=1$$

$$x+y=3$$

$$u=1$$

$$u=3$$

$$v=0$$

$$v=1$$

$$u+v = x+y+x-y$$

$$2x = u+v$$

$$x = \frac{u+v}{2}$$

$$u-v = x+y-x+y$$

$$2y = u-v$$

$$y = \frac{u-v}{2}$$

$$\frac{d(x, y)}{d(u, v)} = \begin{vmatrix} \frac{dx}{du} & \frac{dx}{dv} \\ \frac{dy}{du} & \frac{dy}{dv} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} \end{vmatrix}$$

$$= -\frac{1}{4} - \frac{1}{4} = -\frac{1}{2}$$

from the equation of —

$$\iint_R f(x, y) dA_{xy} = \iint_S f(x(u, v), y(u, v)) \left| \frac{d(x, y)}{d(u, v)} \right| dA_{uv}$$

$$\iint_R \frac{x-y}{x+y} dA = \iint_S \frac{u}{v} \left| \frac{d(x, y)}{d(u, v)} \right| dA_{uv}$$

$$= \iint_S \frac{u}{v} \left| -\frac{1}{2} \right| dA_{uv}$$

$$= \frac{1}{2} \int_0^1 \int_1^3 \frac{u}{v} du dv$$

$$= \frac{1}{2} \int_0^1 \sqrt{\ln(u)}^3 du$$

$$= \frac{1}{2} \ln 3 \int_0^1 \sqrt{v} dv = \frac{1}{4} \ln 3$$

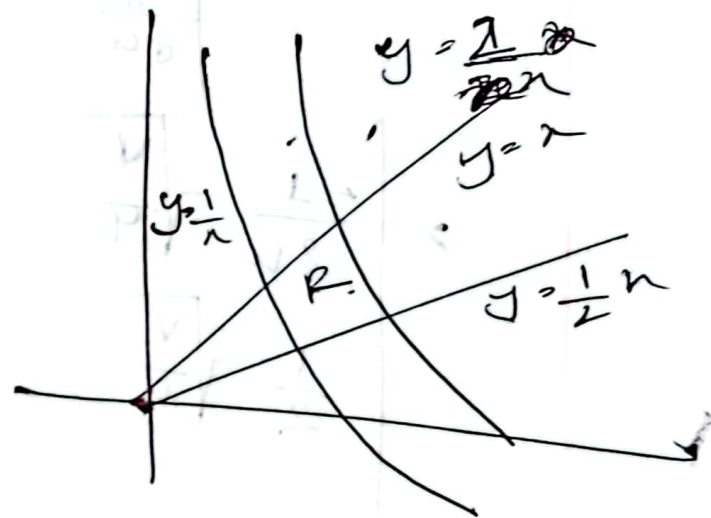
[3] $\iint_R e^{xy} dA$

$$y = \frac{2}{x}$$

$$y = \frac{1}{x}$$

$$y = x$$

$$y = \frac{x}{2}$$



$$\frac{y}{x} = \frac{1}{2}$$

$$\frac{y}{x} = 1$$

$$xy = 1$$

$$xy = 2$$

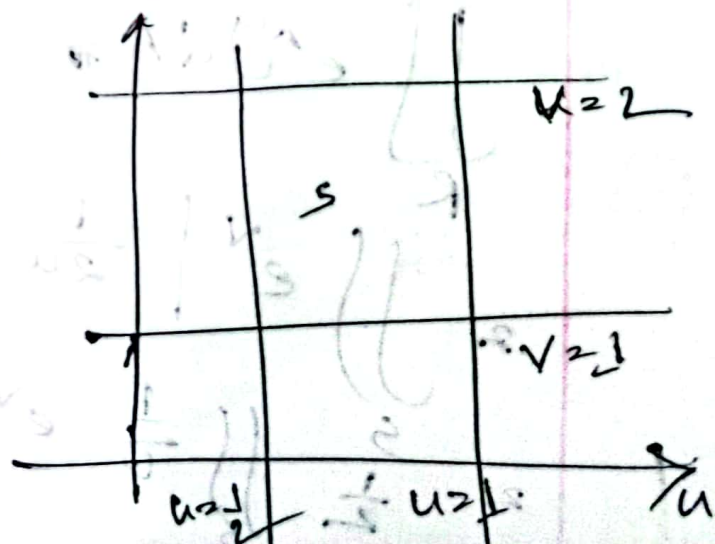
From this transformation the boundary curves in the xy plane and u, v curves in the corresponding lines

$$u = \frac{1}{2}$$

$$u = 1$$

$$v = 1$$

$$v = 2$$



$$x = \sqrt{\frac{u}{v}} \quad y = \sqrt{uv}$$

$$\text{Now, } \frac{d(x, y)}{d(u, v)} = \begin{vmatrix} \frac{dx}{du} & \frac{dx}{dv} \\ \frac{dy}{du} & \frac{dy}{dv} \end{vmatrix}$$

$$= \begin{vmatrix} -\frac{1}{2u} \sqrt{\frac{u}{v}} & \frac{1}{2\sqrt{uv}} \\ \frac{1}{2} \sqrt{\frac{v}{u}} & \frac{1}{2} \sqrt{\frac{u}{v}} \end{vmatrix}$$

$$= -\frac{1}{4u} - \frac{1}{4u} = -\frac{1}{2u}$$

From the formula

$$\iint_R e^{xy} dA$$

$$= \iint_S e^v \left| -\frac{1}{2u} \right| dA_{uv}$$

$$= \frac{1}{2} \iint_S \frac{1}{u} e^v dA_{uv}$$

$$= \frac{1}{2} \int_1^2 \int_{\frac{1}{2}}^4 \frac{1}{u} e^v du dv$$

$$= \frac{1}{2} \int_1^2 e^v \ln[u] \Big|_{\frac{1}{2}}^4 dv$$

$$= \frac{1}{2} \ln 2 \int_1^2 e^v dv = \frac{1}{2} (e^2 - e) \ln 2$$

Example 4:-

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

$$V = \iiint_G dv$$

$$x = au$$

$$y = bv$$

$$z = cw$$

$$\frac{x}{a} = u \Rightarrow \frac{1}{a} = \frac{du}{dx} \quad \text{--- (1)}$$

$$\frac{y}{b} = v \Rightarrow \frac{1}{b} = \frac{dv}{dy} \quad \text{--- (2)}$$

$$\frac{z}{c} = w \Rightarrow \frac{1}{c} = \frac{dw}{dz} \quad \text{--- (3)}$$

$$\therefore \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

$$u^2 + v^2 + w^2 = 1$$

The region S in uvw space

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

$$u^2 + v^2 + w^2 = 1$$

$$\frac{d(x, y, z)}{d(u, v, w)} = \begin{vmatrix} \frac{dx}{du} & \frac{dx}{dv} & \frac{dx}{dw} \\ \frac{dy}{du} & \frac{dy}{dv} & \frac{dy}{dw} \\ \frac{dz}{du} & \frac{dz}{dv} & \frac{dz}{dw} \end{vmatrix}$$

$$= \begin{vmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{vmatrix}$$

[placing (i), (ii), (iii)]

$$= abc$$

③

$$X = \cancel{u \cos u} \sin u + \cos v$$

$$y = \sin v - \cos u$$

$$\frac{d(x, y)}{d(u, v)} = \begin{vmatrix} \frac{dx}{du} & \frac{dx}{dv} \\ \frac{dy}{du} & \frac{dy}{dv} \end{vmatrix}$$

$$\frac{dx}{du} = \frac{d}{du} (\sin u + \cos v)$$

$$= \cos u$$

$$\frac{dx}{dv} = \frac{d}{dv} (\sin u + \cos v) = -\sin v$$

$$\frac{dy}{du} = \frac{d}{du} (\sin v - \cos u) = \sin u$$

$$\frac{dy}{dv} = \frac{d}{dv} (\sin v - \cos u) = \cos v$$

$$\begin{aligned} \therefore \frac{d(x, y)}{d(u, v)} &= \left(\frac{dx}{du} \cdot \frac{dy}{dv} \right) - \left(\frac{dx}{dv} \cdot \frac{dy}{du} \right) \\ &= \cos u \cdot \cos v - \sin u (-\sin v) \\ &= \cos u \cos v + \sin u \sin v \\ &= \cos(u - v) \end{aligned}$$

⑦

$$u = x^2 - y^2$$

$$v = x^2 + y^2$$

$$u + v = 2x^2$$

$$x^2 = \frac{u+v}{2}$$

~~$$x^2 = \frac{u+v}{2}$$~~

$$x = \frac{\sqrt{u+v}}{2}$$

①

$$u - v = x^2 - y^2 - x^2 - y^2$$

$$u - v = -2y^2$$

$$y^2 = \frac{u-v}{-2}$$

$$y^2 = \frac{v-u}{2}$$

$$y = \frac{\sqrt{v-u}}{\sqrt{2}}$$

$$\frac{d(x, y)}{d(u, v)} = \begin{vmatrix} \frac{dx}{du} & \frac{dx}{dv} \\ \frac{dy}{du} & \frac{dy}{dv} \end{vmatrix}$$

$$\frac{dx}{du} = \frac{1}{2\sqrt{2}\sqrt{u+v}} \quad \frac{dx}{dv} = \frac{1}{2\sqrt{2}\sqrt{u+v}}$$

$$\frac{dy}{du} = \frac{1}{2\sqrt{2}\sqrt{v-u}} \quad \frac{dy}{dv} = \frac{1}{2\sqrt{2}\sqrt{v-u}}$$

$$\begin{vmatrix} \frac{1}{2\sqrt{2}\sqrt{u+v}} & \frac{1}{2\sqrt{2}\sqrt{u+v}} \\ -\frac{1}{2\sqrt{2}\sqrt{v-u}} & \frac{1}{2\sqrt{2}\sqrt{v-u}} \end{vmatrix}$$

$$= \left(\frac{1}{8\sqrt{v^2-u^2}} + \frac{1}{8\sqrt{v^2-u^2}} \right) \therefore$$

$$= \frac{1}{4\sqrt{v^2-u^2}}$$

⑧ $u = xy$
 $v = xy^3 \Rightarrow J = ?$

$$m = \frac{u}{y} \quad \text{--- (1)}$$

$$\dot{m} = \frac{v}{y^3} \quad \text{--- (2)}$$

$$\frac{u}{y} = \frac{v}{y^3}$$

$$y = \sqrt{\frac{v}{u}}$$

From eq (1) $\Rightarrow$

$$x = \frac{u\sqrt{y}}{\sqrt{u}} = \frac{u^{3/2}}{y^{1/2}}$$

$$\frac{d(x,y)}{d(u,v)} = \begin{vmatrix} \frac{dx}{du} & \frac{dx}{dv} \\ \frac{dy}{du} & \frac{dy}{dv} \end{vmatrix}$$

$$\therefore \left(\frac{3}{4v} - \frac{1}{4v} \right) = \frac{4}{2v} \quad (\text{Ans})$$

(2) $u = x - 2y$
 $v = 2x + y$

$$\iint_R \frac{x-y}{2x+y} dA$$

$$\begin{cases} x - 2y = 1 \\ x - 2y = 4 \end{cases}$$

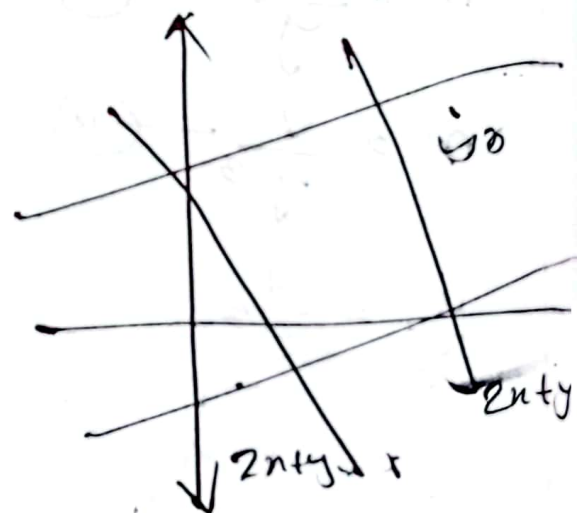
$$\begin{cases} 2x + y = 1 \\ 2x + y = 3 \end{cases}$$

$$u = x - 2y$$

$$v = 2x + y$$

$$u + 2v = x + 4x$$

$$x = \frac{u + 2v}{5}$$



(1) $-2 \times (1) \Rightarrow$

$$v - 2u = y + 4y$$

$$5y = v - 2u$$

$$y = \frac{v-2u}{5}$$

$$u = 1, u = 4$$

$$v = 1, v = 3$$

$$J = \frac{d(x, y)}{d(u, v)} = \begin{vmatrix} \frac{dx}{du} & \frac{dx}{dv} \\ \frac{dy}{du} & \frac{dy}{dv} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{1}{5} & \frac{2}{5} \\ -\frac{2}{5} & \frac{1}{5} \end{vmatrix} = \frac{1}{25} + \frac{4}{25} = \frac{1}{5}$$

$$\iint_R \frac{x-2y}{x+y} dA = \int_1^3 \int_1^4 \frac{u}{v} \left| \frac{1}{5} \right| du dv$$

$$= \frac{1}{5} \int_1^3 \frac{1}{2v} [u^2]_1^4 dv$$

$$= \frac{1}{10} \int_1^3 \frac{1}{v} (4^2 - 1) dv$$

$$= \frac{3}{2} \int_1^3 \frac{1}{v} dv$$

$$= \frac{3}{2} \ln(3)$$

②② $u = x+y$

$v = x-y$

$\iint_R (x-y) e^{x^2-y^2} dA$ over the rectangular

Region R enclosed by the lines $x+y=0$

$x+y=1$, $x-y=1$, $x-y=4$

here, $u = x+y$ — ①

$v = x-y$ — ②

① + ②

$x = \frac{u+v}{2}$

① - ②

$y = \frac{u-v}{2}$

Here,

$u \geq 0$

$v \geq 1$

$u \leq 1$

$v \leq 4$

$$J = \frac{d(x, y)}{d(u, v)} = \begin{vmatrix} \frac{dx}{du} & \frac{dx}{dv} \\ \frac{dy}{du} & \frac{dy}{dv} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{vmatrix}$$

$$= -\frac{1}{4} - \frac{1}{4} = -\frac{1}{2}$$

$$\therefore \iint_R (x-y) e^{x^2-y^2} dA$$

$$= \iint_R (x-y) e^{(x-y)(x+y)} dA$$

$$= \int_0^1 \int_0^1 \sqrt{e^{uv}} \left| -\frac{1}{2} \right| du dv$$

$$= \int_0^1 \left[\frac{1}{2} \sqrt{e^{uv}} \int_0^1 e^{uv} du \right] dv$$

$$= \int_0^1 \frac{v}{2} \left[\frac{e^{uv}}{v} \right]_0^1 dv$$

$$= \frac{1}{2} \int_1^4 (e^v - 1) \frac{dv}{v} = \frac{(e^v) \ln v - v}{(v \ln v) \ln v} = \dots$$

$$= \frac{1}{2} [e^v - v]_1^4$$

$$= \frac{1}{2} (e^4 - 4 - e + 1) = \frac{1}{2} (e^4 - e - 3) \quad (\text{Ans}).$$

$$\frac{1}{x} = \frac{1}{x} - \frac{1}{x} = 0$$

$$A \ln \left(\frac{e^x - 1}{e^x} \right) = \dots$$

$$\ln \left(\frac{e^x - 1}{e^x} \right) = \ln(e^x - 1) - \ln(e^x) = \dots$$

$$v \ln v \left| \frac{1}{v} - \ln v \right| = \dots$$

$$v \ln v \left[\frac{1}{v} - \ln v \right] = \dots$$

$$v \ln v \left[\frac{1}{v} - \ln v \right] = \dots$$